

Code-Layout, Readability and Reusability

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Code follows consistent 4-space indentation for better readability.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into templates, static, uploads, reports, and model files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as <code>ApplicantIncome</code> , <code>LoanAmount</code> , <code>CreditHistory</code> , and <code>prediction</code> clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>home()</code> , <code>predict()</code> , <code>submit()</code> , and dashboard-related methods are clearly named.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are added to explain preprocessing, model loading, prediction workflow, and routing.
6	Modular Design	Code is split into reusable functions/modules	Yes	Frontend, backend, Machine Learning model, CSS, JavaScript, and templates are separated into independent modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code was eliminated and reusable functions were implemented wherever possible.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling improve application reliability and user experience.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Flask Routing Module	Python (Flask)	Handles navigation between Home, Prediction, Dashboard, and Report pages	High
2	Machine Learning Prediction Module	Scikit-learn (Random Forest)	Loan eligibility prediction	High
3	Feature Scaling Module	Scikit-learn	Data preprocessing before prediction	High
4	EMI Calculator Module	JavaScript	Prediction page and dashboard	High
5	PDF Report Generator	JavaScript / HTML	Applicant report generation	Medium

Overall Code Quality Assessment

Aspect	Rating (1 –5)	Comments
Code Layout & Structure	5	Well-structured project with separate frontend, backend, Machine Learning, and static resource directories.
Readability	5	Uses meaningful variable names, modular functions, and consistent coding conventions.
Reusability	5	Modular architecture allows easy reuse of prediction engine, dashboard components, and utility modules.
Documentation / Comments	5	Source code includes comments and follows standard naming conventions for easy maintenance.
Overall Score	5 / 5	The project demonstrates a clean, modular, maintainable, and scalable codebase suitable for future enhancements and deployment.